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# Recovery of Kolmogorov Statistics in the Troposphere

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## ABSTRACT

We study tropospheric aerothermal probe data by using the orthogonal Haar wavelet averages of the dissipation to segment the data. Segmenting the data in this way allows us to isolate regions of distinct mean dissipation. We then use the Haar wavelet transform to derive spectra and structure functions for the segmented regions, thereby recovering Kolmogorov statistics. We also coment on wavelet derived structure functions and point out data anomolies only visible in the wavelet domain.

**Keywords:** geophysical turbulence, dissipation, wavelets

## 1. INTRODUCTION

In the study of optical propagation in the atmosphere the atmospheric temperature variations are often assumed to have a Kolmogorov spectrum, i.e. a spectrum with a $-5/3$ power law. The original derivation of Kolmogorov, the K41 theory, was for the velocity spectrum in a homogeneous, isotropic, random field where constant dissipation was assumed to exist. Although these conditions do not exist over large ranges in the free atmosphere, the assumption of a Kolmogorov inertial subrange has proved very useful in optical propagation studies.

The Air Force is currently gathering extensive upper atmospheric turbulence data to provide input for the design of optical control systems for the Airborne Laser program. This aerothermal probe data has traditionally been processed in the frequency domain, where certain noise sources could be avoided. The processing assumes a power law behavior for the temperature fluctuations and then finds the best power law fit to the spectrum of the data in log-log space. Although the spectra derived from this bulk processing technique has been shown to provide good input for optical propagation studies, it is never-the-less of interest to explore the applicability of certain fluid dynamic assumptions associated with the K41 theory as it applies to this data. This paper contributes to that exploration by considering a heuristic data segmentation based on regions of relativity constant dissipation. The segmented data is then investigated for agreement with the K41 theory. In particular we ask to what extent the $p$th order structure functions associated with the data agree with the K41 $p/3$ power law prediction. It is found that if we restrict the computations to regions of relativity constant dissipation the $p/3$ power can be recovered out to high order in $p$. Several side benefits of the decomposition are the isolation of turbulent bursts consistent with the blini or pancake model of tropospheric turbulence, and the identification of an otherwise undetected instrumentation noise.

We focus on a section of Air Force data taken in the Fall of 1997 at 40,000 feet altitude over New Mexico. The data represents high-resolution measurements of very high Reynolds number flows with a variety of turbulence intensities. It is subject to various external effects, including instrumentation errors, that are not obvious in the space or Fourier domains, but are quite clear in the wavelet domain. Finally, the data contains both *intrinsic* intermittency associated with turbulent bursts and *outer scale* intermittency associated with large regions of the atmosphere having disparate turbulent statistics.

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## 2. SOME ASPECTS OF ATMOSPHERIC TURBULENCE

Kolmogorov's original works, often referred to as K41 and K61,[2–4] are the most substantial and predictive description of turbulence from first principles through the Karman-Howarth-Monin relation. Overviews of Kolmogorov's assumptions and its implications can be found in[5–8] and many others. While these ideas only apply to kinetic energy dynamics, velocity increments, power spectral densities and related quantities, Corrsin and Oboukhov independently extended universal equilibrium concepts to thermal fluctuations.[9,10] This extension was later confirmed experimentally by Gibson and Schwarz.[11] In this article, we analyze thermal data in the absence of any information about the kinetic energy of the flow.

A consequence of Kolmogorov's assumptions, as extended by Corrsin and Oboukhov, is that the temperature power spectral density and structure functions obey the following power laws within the inertial range:

$$E(k) = Ck^{-5/3} \tag{1}$$

$$T_p(r) = \langle |\delta\theta(\mathbf{x}, \mathbf{r})|^p \rangle = C_p r^{p/3} \tag{2}$$

where $\delta\theta(x, r) \equiv \theta(x + r) - \theta(x)$

Of course, the theorems of Kolmogorov and others are based on the existence of a universal equilibrium and an inertial-dissipative range, and so their applicability to measurements of thermal fluctuations in the troposphere is severely limited by the ability to find regions of fluid flows where these conditions exist. The problem is that neither kinetic energy dissipation nor thermal dissipation remains constant over large ranges in the troposphere. Intermittent bursts, characterized by regions of relatively intense dissipation, play a central role. These intermittent bursts cause departures from $p/3$ scaling. However, there is strong evidence that in the troposhphere there are distinct pancake-like regions, each having their own turbulent statistics. While universal equilibrium exists within the pancakes, the kinetic or thermal dissipations vary from one pancake to the next. We use wavelet averages to identify regions of relatively constant thermal dissipation, to isolate localized bursts, and thus to segment the data into regions where Kolmogorov's universal equilibrium is likely to apply.

## 3. THE DATA

### 3.1. Data Collection and Recording

The data, section 97223-04 of the Air Force data set, is shown in figure 1. It was collected by an aerothermal probe mounted under the nose of a C-135E aircraft flying at 40,000 ft. over New Mexico in the Fall of 1997. The probe measured wire resistance changes induced by wire temperature changes resulting from atmospheric temperature

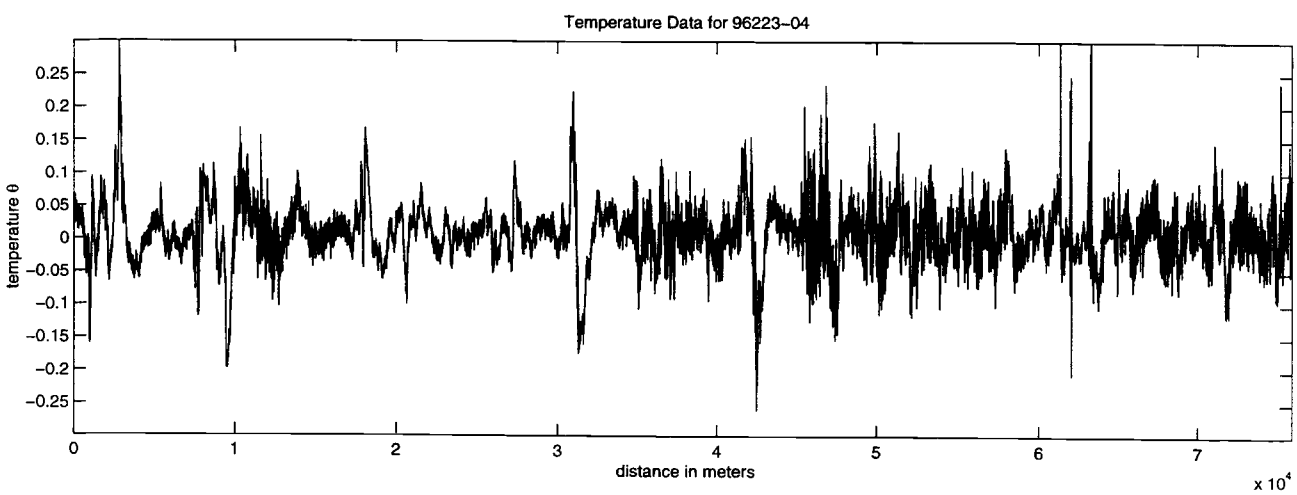


**Figure 1.** Temperature variations for 97123-04. The data was high pass filtered at .01 Hz



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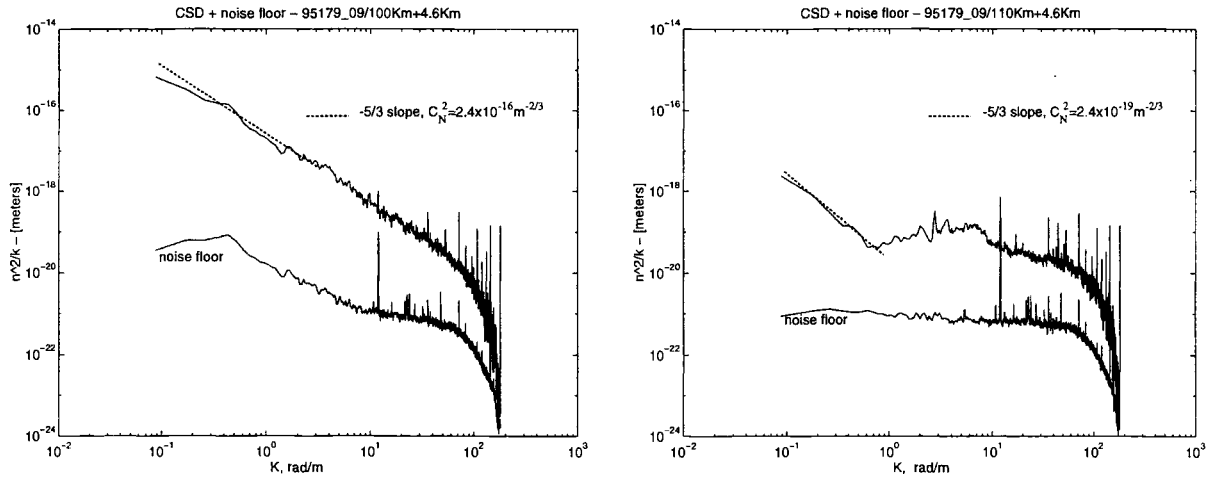


**Figure 2.** Power spectral densities of thermal fluctuations in strong (left) and weak (right) turbulence. The electronic noise floor is shown on each figure. In weak turbulence, an additional noise bump is evident in the data.

and velocity fluctuations. The probe is operated in a low overheat mode which makes it much more sensitive to temperature fluctuations than to velocity fluctuations, which are ignored.

The electronics and data recording system support a 12 K-Hz sample rate. This corresponds to a data sample every 1.8 cm. However the probe had a thermal bandwidth of approximatle 700 Hz (3 db point) which corresponds to a wave number of $k < 20\ m^{-1}$ and a wavelength of $\lambda > 30$ cm. There was a high-pass filter at .1 Hz to separate AC and DC channels. This invokes the additional limits $k > .003m^{-1}$ and $\lambda < 2.2km$.

### 3.2. Noise

A typical power spectral density (PSD) taken in strong turbulence is presented on the left in Fig. 2. This PSD was derived from a temperature time series (not 97223-04) and is presented in index of refraction units. The aircraft velocity, 220 m/sec, was used to transform the abscissa to units of radians/meter ($m^{-1}$).[12,13] The PSD exhibits the characteristic Kolmogorov -5/3 slope out to the sensor roll-off. Note that the signal lies well above the measured electronic noise floor. The 400 Hz spike in the electronic noise floor (11 $m^{-1}$) in this figure is attributed to the aircraft power system. Most of the other spikes are harmonics of this frequency.

The right graph in Fig. 2 shows a similar PSD, but in weak turbulence. Here we see an additional feature, which looks like noise but lies well above the known electronic noise floor. This apparent noise been identified as acoustically driven thermal variations. The acoustical source is an engine noise phenomenon known as jet screech.[14] In weak turbulence this noise becomes a factor in data quality. It effectively limits the observable wavenumbers to $\kappa < .5m^{-1}$ and observable wavelengths to $\lambda > 12m$. In low turbulence situations these limits must be added to those due to system bandwidths described above.

### 3.3. Conventional Processing

Because of sensor roll-off and the various noise sources, the preferred method for bulk data processing has been in the frequency domain. Following standard procedures,[15] the data is partitioned into 4.8 Km segments which are further subdivided into sixteen 300-m subsegments. A PSD is calculated for each of the subsegments and the subsegment PSDs are averaged to get the PSD for the 4.8 Km segment. The segment PSD is then fit to a power law over a frequency regime which avoids the obvious noise features. For weak turbulence the power law is fit in the region between .05 and .5 $m^{-1}$, whereas in strong turbulence the fit is between .5 and 10 $m^{-1}$. The atmospheric characteristics, $C_n^2$ and $\alpha$, are read off of the resultant power law.

$$E(k) = 0.2488\ C_n^2\ k^{\alpha}, \tag{3}$$

where the factor 0.2488 is a conversion from index of refraction to thermal quantities. $C_n^2$ and $\alpha$ for 97223-04 is shown in figure 5, where it is superimposed on the dissipation.



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## 4. WAVELETS, SPECTRA, AND STRUCTURE FUNCTIONS

The Haar wavelet coefficients for a signal $\{a_0(n) : 0 \leq n < 2^M\}$ are given by:

$$\begin{aligned} a_m(n) &\equiv \frac{1}{2}\left[a_{m-1}(2n) + a_{m-1}(2n+1)\right], \quad 0 \leq n < 2^{M-m} \\ d_m(n) &\equiv \frac{1}{2}\left[a_{m-1}(2n) - a_{m-1}(2n+1)\right], \quad 0 \leq n < 2^{M-m}. \end{aligned} \tag{4}$$

In the wavelet literature, the numbers $a_m(n)$ are called the $m$-th level *Haar average coefficients* and $d_m(n)$ the $m$-th level *Haar detail (wavelet) coefficients.* Note that the normalization above differs from that found in most of the literature, where the factors in equation (4) are taken to be $1/\sqrt{2}$ instead of $1/2$. That makes the associated Haar wavelets orthonormal instead of merely orthogonal, as here. The standard normalization is mathematically convenient but results in unnecessary computational complexity. Division by 2, as above, merely amounts to a shift in the binary representation, which renders the fast Haar transform (FHT) algorithm of order $O(log(n))$.

Summation of the detail coefficients at a scale m results in the power at scale m. Thus one can define the scale spectrum analogous to the Fourier spectrum of a signal (see Papanicolaou, et. al.[16] in these proceedings). The spectra of figures 7 and 8 were calculated in this way. Note the characteristic smoothness of these spectra in relation to the Fourier spectra of figures 2. The smoothness arises due to wavelet averaging, which is always just right for the scale.

One might think that such averaging would also be valuable in calculating structure functions. In fact this idea has been used to some advantage in Katul, et. al..[17] The argument goes roughly as follows: Let $r \to 2^{m-1}b$ and $t \to 2n \cdot 2^{m-1}b$, where b is the separation between data samples and $1 \leq m \leq M$ and $0 \leq n < 2^{M-m}$. Assuming ergodicity, the ensemble average can be replaced by the time average, provided a large number of samples is taken. At the separation $r = 2^{m-1}b$, details at finer scales ($m' < m-1$) are expected to average out, so $\theta(t)$ can be replaced by $a_{m-1}(2n)$. Therefore

$$|\delta\theta(t,r)|^p = |\theta(t+r) - \theta(t)|^p \to |a_{m-1}(2n+1) - a_{m-1}(2n)|^p = 2^p|d_m(n)|^p.$$

Taking the time average by summing over $n$ gives

$$T_p(2^m b) = \langle|\delta\theta(t,2^m b)|^p>\rangle \sim 2^{p+m-M} \sum_{n=0}^{2^{M-m}-1} |d_m(n)|^p \equiv T_p^d(2^m b). \tag{5}$$

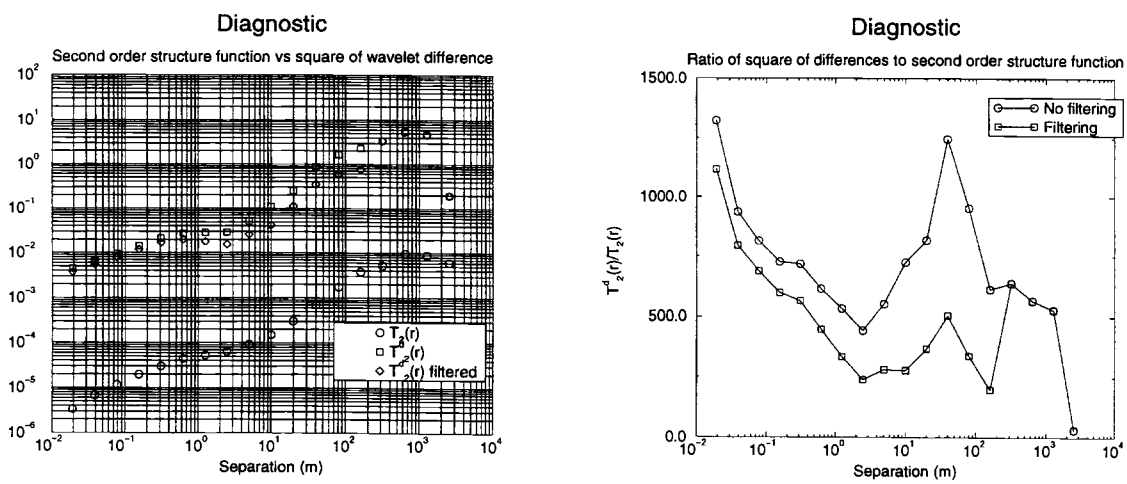


**Figure 3.** Second order structure functions $T_2(r)$ and powers of fast Haar wavelet coefficients, $T_2^d(r)$, are calculated to see if they are proportional. Averages are calculated over a distance of approximately 5,200 m, well beyond the integral scale of this flow. On the left, one can see that they share the same general trend, but on the right, they are seen to be far from proportional to one another. Filtering refers to the prescription of Katul et al[17]with filtering parameter $\delta = 5.0$, wherein large deviation wavelet details are discarded.



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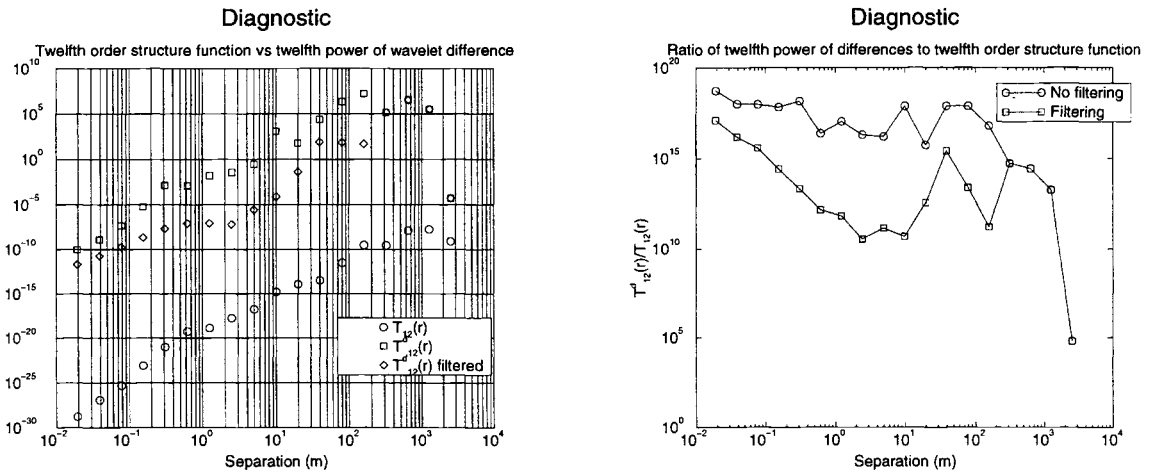


**Figure 4.** Twelfth order structure functions $T_{12}(r)$ and powers of fast Haar wavelet coefficients, $T_{12}^d(r)$, are calculated to see if they are proportional. Note that the plot on the right has a logarithmic vertical scale. For higher-order structure functions, one can see that even the general trend is different, and the "constant" of proportionality varies by several orders of magnitude. Filtering is performed as described in Fig. 3.

The right side is, by definition, the *discrete structure function of order* $p$ as computed by the FHT.

In Figures 3 and 4, we have computed $T_p(2^m b)$ for $p = 2$ and $p = 12$ directly from the raw data, and compared this with $T_p^d(2^m b)$ as computed from 5 with the FHT. Although the two are similar, they exhibit sufficient differences to conclude that the approximation (5) is rather crude. Of course, the argument above is merely heuristic, so there is no contradiction.

One can calculate the exact relationship between $T_p(2^m b)$ and $T_p^d(2^m b)$ by back substituting in equation 5 to write $T_p^d$ completely in terms of $a_0(n)$, i.e. directly in terms of the data, and comparing it time average for $T_p$ in terms of the data. It is immediately clear that $T_p^d$ contains many additional cross terms than $T_p$, and thus the differences between $T_p^d$ and $T_p$ are signal dependent.

A more heuristic view notes that if the ensemble average in 2 is replaced by a time average, then $T_p(2^m b)$ is a time average of the $p$-th power of the absolute fluctuation $|\delta\theta(t,r)|$. On the other hand, $T_p^d(2^m b)$ is a *coarse-grained* time average (at $\Delta t = 2^m b$) of the $p$-th power of the absolute fluctuation of a *blurred (to* $\Delta t = 2^{m-1} b$*) version* of $\theta(t)$. There is no *a priori* reason why these two sets of operations should give the same result.

Given these considerations we have chosen to calculate structure functions directly from the data rather than from the wavelet coefficients. This is not a detriment, since structure functions are inherently smoother than Fourier spectra.

## 5. RESULTS

If one measures spectra and structure functions without regard to dissipation, one finds that they do not obey the K41 power laws. In particular the $p/3$ scaling of the *pth* order structure function does not hold. However, calculating the same functions along regions that have constant dissipation, one recovers Kolmogorov scalings.

A second and unforeseen advantage of our analysis is the discovery of regular patterns buried within the thermal measurements. Whether or not the source of these patterns is natural or artificial is still under investigation, but the fact remains that their existence was invisible to both the time domain and frequency domain analysis.

### 5.1. Recovery of Kolmogorov Scaling

We segment the data into regions of uniform dissipation by examining the wavelet decomposition. While we do not have access to all three components of dissipation, we can calculate $\chi = (\frac{d\theta}{dx})^2$. If the flow is locally isotropic within the resolution of our measurements (a reasonable assumption), and we average over small distances, then we are



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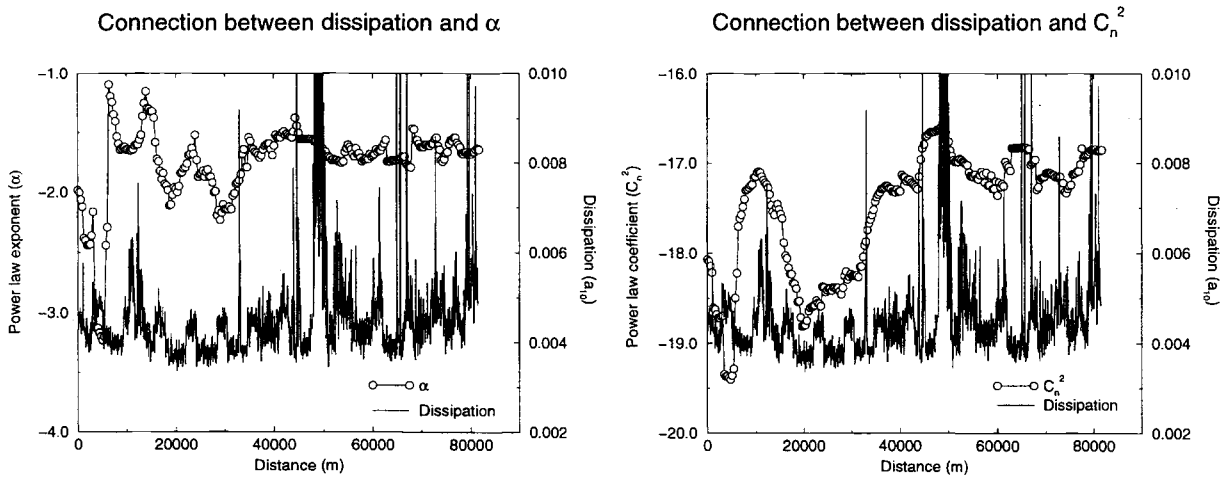


**Figure 5.** Dissipation, or $\chi$, for 97223-04, superimposed on $\alpha$ on the left and $C_n^2$ on the right. $C_n^2$ and $\alpha$ are calculated as described in 3.3. One can see that the sharp transitions in $\alpha$ and $C_n^2$ correspond roughly to sharp transitions in dissipation.

assured that $\chi$ is proportional to the thermal dissipation. In our analysis we approximate the dissipation, or $\chi$, by taking centered differences of the raw temperature data. The results of this calculation is shown in figure 5.

Levels 11 through 15 of the Haar wavelet decomposition of dissipation are shown in figure 6. One is immediately impressed with the fact that the dissipation is not constant even at coarse resolutions. However, within the entire 80 kilometer data set, there are several regions wherein the dissipation appears to be roughly constant. For instance, the first 45 kilometers of the thermal dissipation exhibit a thermal dissipation is somewhere between $4 \times 10^{-3}$ and $5.1 \times 10^{-3}$. Afterward, there are localized regions where the dissipation surges several times higher, depending upon the scale over which one is averaging.

Another interesting feature is that the brief regions of elevated dissipation have substantial structure at small scales, while the regions of constant dissipation have relatively little structure aside from the periodicity discussed in the next section. We can segment the data using the wavelet averages of dissipation to tell us where dissipation is relatively constant and thus where Kolmogorov's universal equilibrium hypotheses are valid. Generally, if one calculates the power spectral density in regions of constant dissipation, one recovers a nearly Kolmogorov spectrum. However, high order structure functions, while following Kolmogorov's $p/3$ scaling in regions of constant dissipation, do not follow such scaling data near localized bursts. Spectra also deviate from Kolomogorov in these regions, but not neary so much as for higher order structure functions. (See figure 7)

Notice that elevated regions of dissipation can occupy several kilometers. While localized, these regions have a substantial scale. Such data is entirely consistent with radar observations of large flat regions in the atmosphere co-existing at the same level. In other words, one explanation for the data is that the aircraft passed through a 2-kilometer region having a mean thermal dissipation different from that in the surrounding atmosphere. If this were the case, we should be able to recover Kolmogorov statistics when we calculate spectra entirely within the burst. Indeed, as seen in Fig. 8, this is precisely the case. Anomolous behavior in spectra can be explained as a lack of frequency resolution for the dominant part of the signal. For this window, one would expect the spectrum to be the sum of two distinct $Ck^{-5/3}$ spectra. The high dissipation portion would dominate the sum because C is related to the cube root of $\chi$. However, one can also see in Fig. 6 that the high-dissipation portion of the signal has relatively small measure, meaning that there is very poor frequency resolution. Therefore, one would expect the spectrum and structure functions to contain a fair amount of noise.

## 5.2. Observation of embedded waves

As discussed in §5, experimental data is seldom perfect. In this case, the thermal data rolls off at small wavenumbers, is regularized by a finite probe response at high wavenumbers, and is contaminated by noise from the aircraft if the



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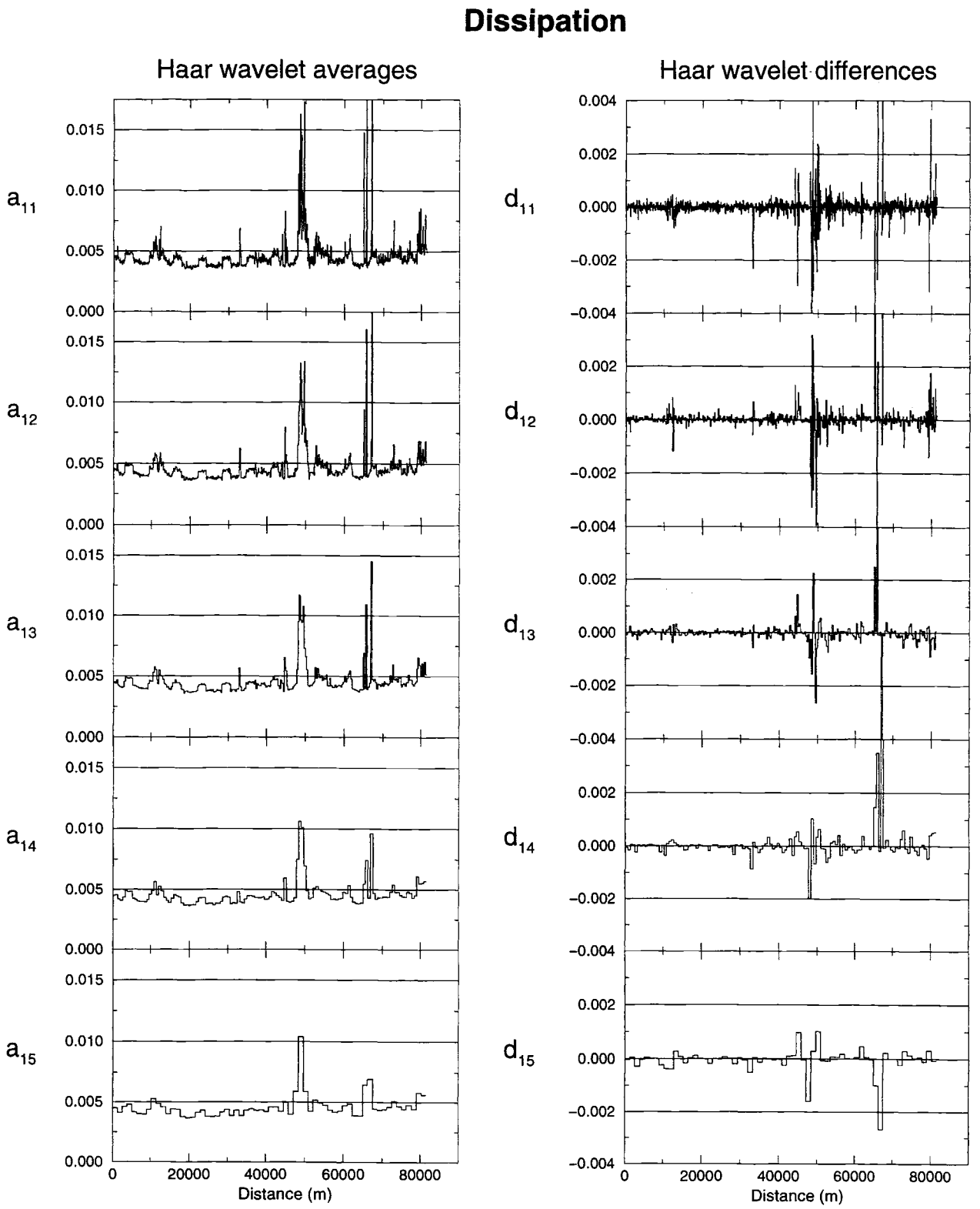


**Figure 6.** Haar averages and details of $\left(\frac{d\theta}{dx}\right)^2 \propto \chi$ for one set of thermal measurements in the troposphere. The averages can be thought of as running averages of the signal, and the details can be expressed as differences of averages at the previous (finer) scale.



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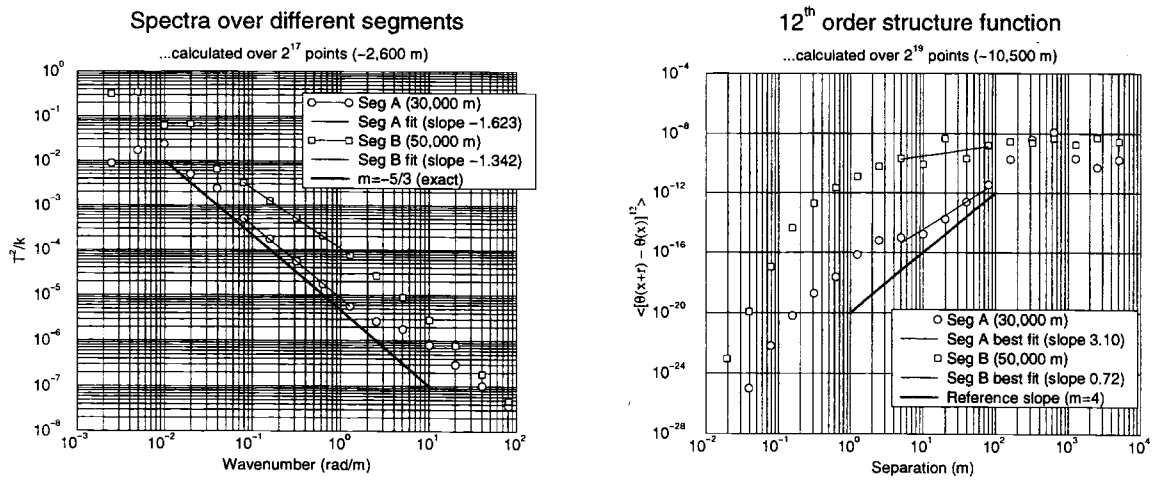


**Figure 7.** Spectrum (left) and $12^{th}$-order structure function (right) for different segments of the signal displayed in Fig. 6. In Segment A (displaced 30,000 m form the origin), the dissipation is roughly constant while in Segment B (displaced 50,000 m from the origin), the dissipation is changing abruptly. A best fit to the data over the range of values not affected by the low-pass roll-off and the aircraft-induced turbulence shows that Kolmogorov spectra are realized in Segment A and not in Segment B. The high-order structure function on the right shows an even sharper difference between the two segments. In fact the left most point in the line fit to Seg A is probably affected by the engine noise, since this region is a region of low turbulence. Disregarding this point makes the Seg A fit of the $12^{th}$ order structure function even closer to $p/4$.

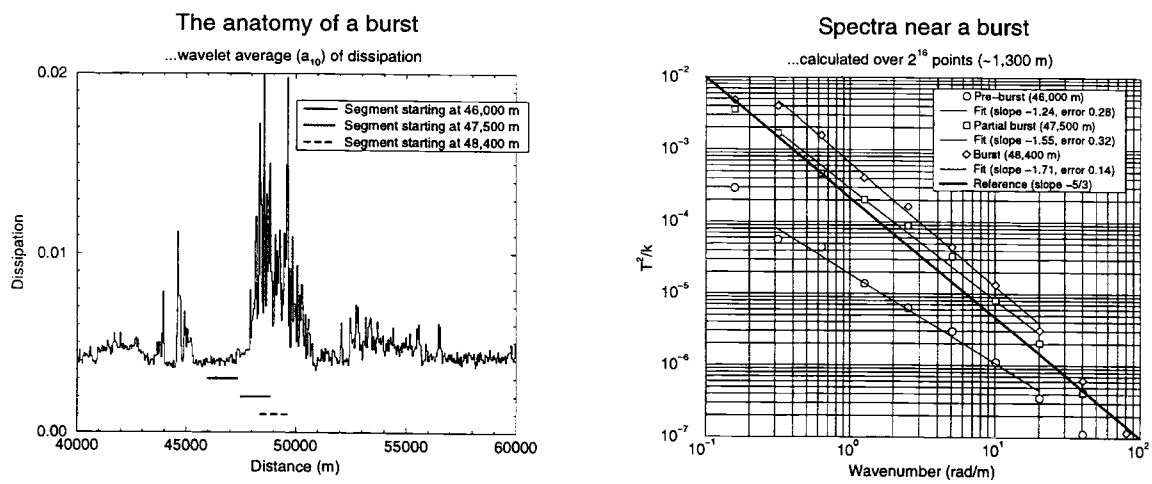


**Figure 8.** Spectra calculated around a burst. On the left, we see three segments near a sharp increase in dissipation. The first is in a region of relatively calm dissipation. The last is entirely within the region of high dissipation. The middle segment straddles the two. The Haar spectra over these three intervals are shown on the right. Notice that the spectrum does not obey a power law in the transition region. Though a best fit is provided, there is considerable variability in the slope over the region of interest.



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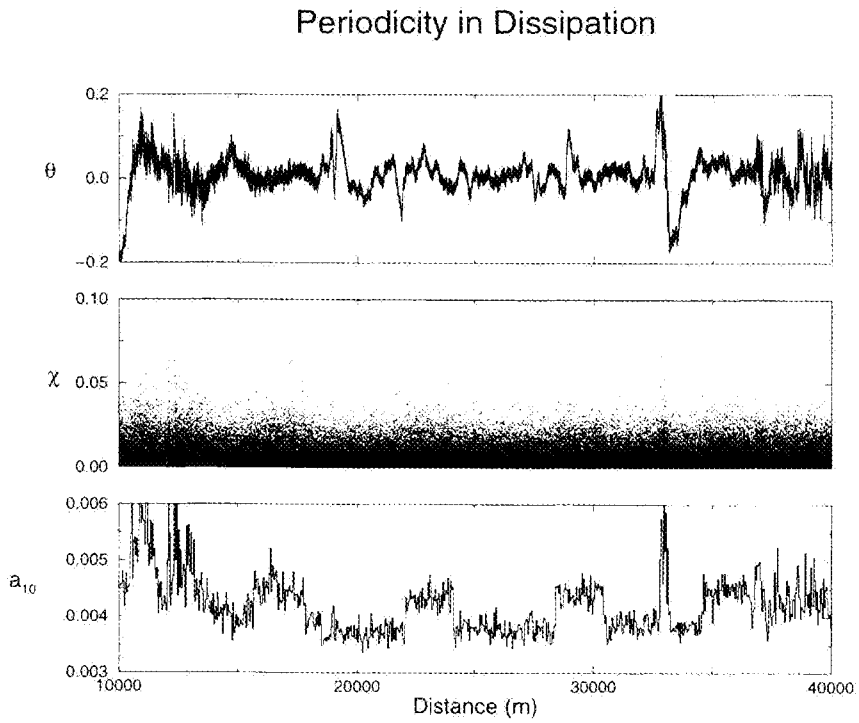


**Figure 9.** Expanded view of embedded square waves in the thermal fluctuations. The embedded waves are not apparent in the downsampled raw temperature $\theta$ shown at top, or in the downsampled raw thermal dissipation $\chi$ shown in the center. (Downsampling keeps the file sizes managable for our plotting package.) However, Haar wavelet averages, $a_{10}$, of the thermal dissipation reveal these waves among the clutter. Note that the vertical scales vary for $\chi$ and $a_{10}$, due to cancellation.

atmospheric turbulence intensities are not high enough. In addition, the wavelet analysis of the thermal dissipation uncovered an unanticipated feature in the data that may be physical or may be instrumental.

Re-examining Fig. 6, we notice a regular pattern in the calm region extending from 13 km up to 30 km. Figure 9 is a blowup of this region showing the tenth wavelet average along with $\theta$ and $\chi$. It appears there is a square wave multiplying the dissipation. This apparent periodicity in $\left(\frac{d\theta}{dx}\right)^2$ means that the *roughness* of $\theta(x)$ has a regular spatial structure, though is says nothing about the magnitudes of the temperature fluctuations themselves. The periodicity is not noticeable in $\theta$ at all, and the periodicity in $\chi$ is masked by the large-scale variations. However, the periodic structure is clearly visible in the tenth wavelet average.

Although we do not have an explanation for this embedded wave, it comprises only about 1 percent of the signal power. Therefore it does not effect the conventional processing used to extract atmospheric parameters, $\alpha$ and $C_n^2$. It is never the less interesting, and is most apparent in the wavelet averages.

## 6. CONCLUSION

We have described a simple and effective method for analyzing real experimental data from turbulent flows. Using fast Haar transforms, we recovered relevant Kolmogorov statistics over segments of turbulent data by examining averages of thermal dissipation at different scales. While segmenting the data is essential to both frequency and scale analysis, our analysis gives local averages which are useful in identifying regions where Kolmogorov's power laws apply. Finding bursts and segmenting the data into regions where the dissipation is roughly constant yields good agreement with the anticipated $k^{-5/3}$ spectral density behavior and the $r^{p/3}$ structure function behavior at order $p$. Moreover, we have found embedded waves in the thermal dissipation that do not appear in the time and frequency domain analyses. While our Haar analysis is simple and begs for enhancements, it has proven to be a robust and effective technique for studying real data.



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## 7. ACKNOWLEDGEMENTS

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